Shell Scripting Lesson

Skills you will learn

Version Control & Linux: Shell Scripting

1. Introduction to Shell Scripting

What is a Shell?

A shell is a command-line interface that enables users to interact with the operating system. The most commonly used shell is **Bash (Bourne Again Shell)**.

What is a Shell Script?

A shell script is a file containing a sequence of shell commands that automate tasks, making them faster and more efficient.

2. Writing Your First Shell Script

1. Open a terminal and create a new script file:

```
nano myscript.sh
```

Copy Explain

2. Add the following content:

```
#!/bin/bash
echo "Hello, welcome to shell scripting!"
```

Copy Explain

3. Save and exit (Press **CTRL+X**, then **Y**, and **Enter**).
4. Make the script executable:

```
chmod +x myscript.sh
```

Copy Explain

5. Run the script:

Ask AI Tutor

```
./myscript.sh
```

Copy Explain

3. Variables and User Input

Defining Variables

- Variables in Bash do not require a data type.
- Syntax: `variable_name="value"`
- Example:

```
#!/bin/bash
name="Alice"
echo "Hello, $name!"
```

Copy Explain

Reading User Input

- Use the `read` command to accept input from the user.
- Example:

```
#!/bin/bash
echo "Enter your name:"
read name
echo "Hello, $name! Welcome to shell scripting."
```

Copy Explain

4. Conditional Statements

Using `if-else` Statements

- Syntax:

```
if [ condition ]; then
    # Code to execute if condition is true
elif [ another_condition ]; then
    # Code to execute if another condition is true
else
    # Code to execute if both conditions are false
fi
```

Copy Explain

- Example:

```
#!/bin/bash
echo "Enter a number:"
read number

if [ $number -gt 0 ]; then
    echo "The number is positive."
```

```
elif [ $number -lt 0 ]; then
    echo "The number is negative."
else
    echo "The number is zero."
fi
```

5. Loops in Shell Scripting

For Loop

```
#!/bin/bash
for i in {1..5}; do
    echo "Number: $i"
done
```

Copy Explain

While Loop

```
#!/bin/bash
counter=1
while [ $counter -le 5 ]; do
    echo "Counter is $counter"
    ((counter++))
done
```

Copy Explain

Exercise: Print Numbers from 1 to 10

1. Create a new script:

```
nano print_numbers.sh
```

Copy Explain

2. Add the following content:

```
#!/bin/bash
for i in {1..10}; do
    echo $i
done
```

Copy Explain

3. Run the script:

```
chmod +x print_numbers.sh
./print_numbers.sh
```

Copy Explain

6. Functions in Shell Scripting

Creating and Using Functions

```bash
#!/bin/bash

greet() {
    echo "Hello, $1!"
}

greet "Alice"
```

Example: Function to Add Two Numbers

```bash
#!/bin/bash

greet() {
    echo "Enter first number:"
    read num1
    echo "Enter second number:"
    read num2
    sum=$((num1 + num2))
    echo "The sum is: $sum"
}

greet
```

7. Practical Example: User Management Script

Create a script that adds a new user and sets a password:

```bash
#!/bin/bash

echo "Enter username to add:"
read username
echo "Enter password:"
read -s password

sudo useradd -m "$username" && echo "$username:$password" | sudo chpasswd
sudo usermod -aG sudo "$username"

echo "User $username created and added to sudo group."
```

Run the script:

```bash
chmod +x add_user.sh
./add_user.sh
```

8. Summary of Commands

Command	Description
`nano filename`	Create/Edit a file
`chmod +x script.sh`	Make script executable
`./script.sh`	Run the script
`read variable_name`	Get user input
`if [ condition ]; then`	Conditional execution
`for i in {1..5}; do ... done`	Loop execution
`function_name() { ... }`	Define a function
`sudo useradd username`	Add a new user

This guide provides fundamental knowledge of **Shell Scripting** and its usage in system automation!

AI TUTOR ACTIONS

Generate a practice quiz Summarize this unit Ask about this unit

Complete unit

Previous

Week 1: Day 5
Guided: Scripting for Remote SSH

Next